Nischal Singh

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Summary

Computer Science graduate with a good foundation in machine learning and deep learning architectures. Skilled in applying point cloud data analysis and computer vision to drive innovation in project development. Seeking a full-time position to leverage technical skills and contribute to real-world applications.

Education

Madhav Institute of Technology and Science, Gwalior — CGPA: 7.8
B.Tech, Computer Science and Engineering — 2020-2024

Experience

Indian Institute of Space Science and Technology, Thiruvananthapura

- **Machine Learning Research Intern:** Researching on utilizing transformer architecture on semantic segmentation of 3D LiDAR point cloud data.
  January - Present, 2024
- **Summer Internship:** Worked on the project "Classification of Multispectral UAV Images Using Segment Anything Model-Based Machine Learning Approaches". Utilized SAM's automated segmentation for labeled ground truth data, benchmarking machine learning models like SVM, Random Forest, and XGBoost.
  May - July, 2023

Projects

smart_cartpole-v1 || *Python, PyTorch, Gymnasium* — Link to Source Code

In this reinforcement learning project, A comparison is done between the performance of Deep Q-Network and Q-learning on the CartPole-V1 environment from OpenAI Gym. The goal is to learn an optimal policy for balancing the pole on a cart.

pointnet2 || *Python, PyTorch, Open3D* — Link to Source Code

This project is a PyTorch implementation of two pivotal papers, PointNet and PointNet++, on utilizing deep learning in point cloud data as a direct point input.

transpear || *Python, PyTorch* — Link to Source Code

This is a transformer model built from scratch, trained on Shakespeare's text. This model's output is random generation of text, but very similar to Shakespeare's writing style.

bigram || *Python, PyTorch, Jupyter* — Link to Source Code

Two bigram models are built from scratch, which are used to predict the next character, given a character. The first model is a statistical model, and the second is a neural network built from scratch.

Technical Skills

Concepts	: Data Structures and Algorithms, OS, Computer Networks, OOPs, DBMS.
Languages	: Python, C++, C, Shell.
Skills	: Research, Computer Vision/Machine Learning, Reinforcement Learning.
Frameworks/Tools	: PyTorch, Tensorflow, Open3D, Pandas, Django, Git, GitHub, Jupyter, VS Code.

Achievements

- Presented the work at ISRS-ISG National Symposium'23 on "Assessing the performance of Segment Anything Model (SAM) based labels for training machine learning models in UAV data classification".
- Solved 300+ questions on various platforms such as Leetcode, GeeksforGeeks,and Codeforces.